

Basics of complex variables

Consider the complex variable y

$$y = a + ib$$

where both a & b are real

$$\operatorname{Re}(y) = a \quad \leftarrow \text{real part}$$

$$\operatorname{Im}(y) = b \quad \leftarrow \text{imaginary part}$$

The "complex conjugate" of y is

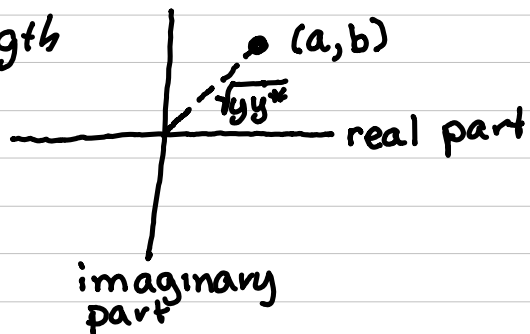
$$y^* = a - ib$$

so:

$$yy^* = [\operatorname{Re}(y)]^2 + [\operatorname{Im}(y)]^2 = a^2 + b^2$$

(remember, $i^2 = -1$)

the squared length
of a vector
in the
"complex
plane"



Exponentials

$$e^{ix} = \cos x + i \sin x$$

$$\operatorname{Re}[e^{ix}] = \cos x$$

$$\operatorname{Im}[e^{ix}] = \sin x$$

$$[e^{ix}][e^{ix}]^* = \cos^2 x + \sin^2 x = 1$$

ck this:

$$e^{ix} e^{-ix} = 1 \checkmark$$

Dirac delta function $\delta(x)$

A continuous distribution such that

$$\delta(x) = \begin{cases} \infty & x = 0 \\ 0 & x \neq 0 \end{cases}$$

$$\underline{\text{and}} \quad \int_{-\infty}^{\infty} \delta(x) dx = 1$$

$$\underline{\text{and}} \quad \int_{-\infty}^{\infty} f(x) \delta(x) dx = f(0)$$

$$\underline{\text{and}} \quad \delta(x-a) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{iq(x-a)} dq$$